

Lecture Slide – 13
Spline Method of
Interpolations

Spline Method of Interpolation

1. What is Interpolation?

2. Why Splines? (Runge)

3. Linear Splines

4. Quadratic Splines — Theory

5. Matrix Setup & Solution

6. Cubic Splines

What is Interpolation?

Definition: Given data points (x_0, y_0) , (x_1, y_1) , ..., (x_n, y_n) , find the value of y at a value of x that is NOT in the table.

Key Concepts

- Polynomials are the most common interpolants — easy to evaluate, differentiate, and integrate
- Known data points are called nodes or knots
- Interpolating function must pass through ALL given data points
- Three common spline types: Linear (C^0), Quadratic (C^1), Cubic (C^2)

Rocket Example — Data Table

Upward velocity of a rocket measured at specific times:

t (s)	v (m/s)
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

Goal: Find velocity at $t = 16$ s (not in table) using spline interpolation.

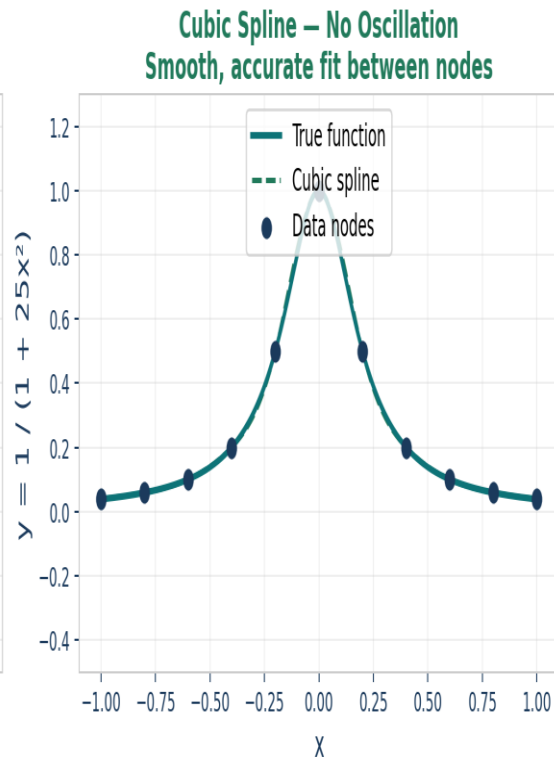
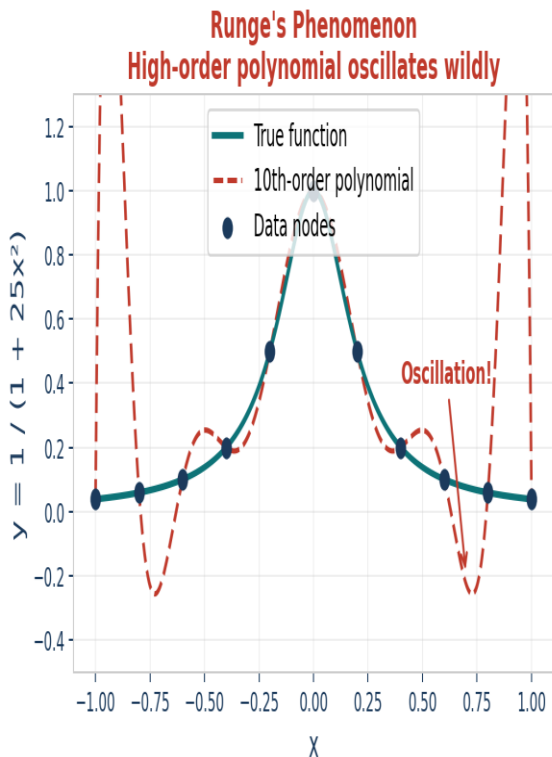
Why Splines? — Drawbacks of High-Order Polynomial Interpolation

High-Order Polynomial Issues

- Runge's Phenomenon: wild oscillations between data points
- Increasing polynomial order does NOT always improve accuracy
- Adding one new data point changes the ENTIRE polynomial
- Error grows rapidly near the endpoints
- Computationally unstable for large n

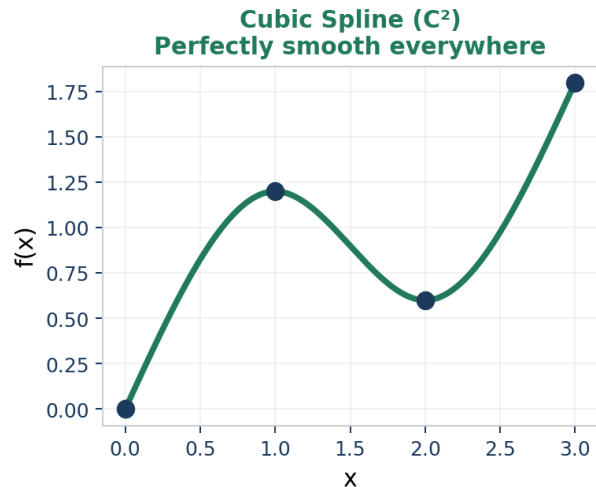
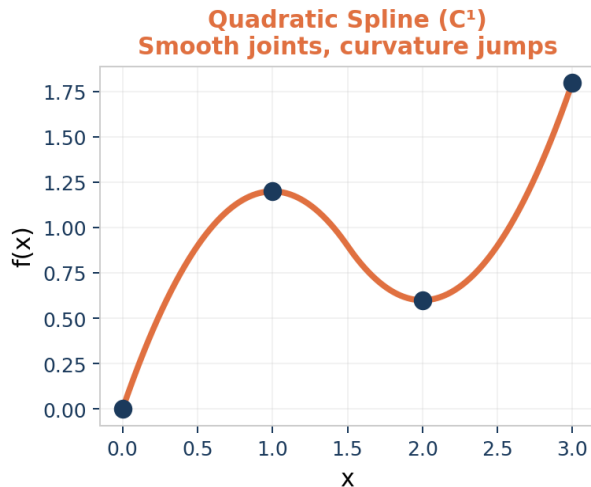
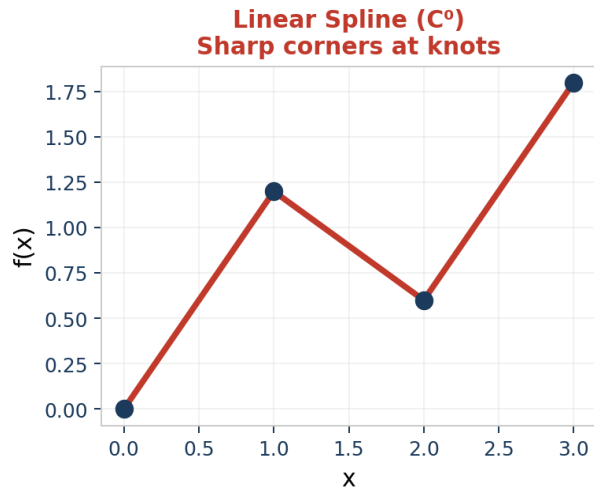
Spline Fix: Use many low-degree polynomials, each fitted locally between two consecutive knots. Join them smoothly.

Why Splines? — Runge's Phenomenon vs Spline Interpolation



Visual Comparison of Spline Types

Visual Comparison of Spline Types at Knot Points



Linear (C^0)

Function continuous at knots. Sharp corners — 1st derivative jumps.

Quadratic (C^1)

Function and 1st derivative continuous. Curvature (2nd deriv) still jumps.

Cubic (C^2)

Function, 1st and 2nd derivative all continuous. Perfectly smooth curve.

Linear Spline Interpolation — Theory & Formula

Concept: Draw straight-line segments between every pair of consecutive data points. For $n+1$ points $\rightarrow$ n linear polynomials.

General Formula

For each interval $x_i \leq x \leq x_{i+1}$ ($i = 0, 1, \dots, n-1$):

$$f(x) = f(x_i) + [(f(x_{i+1}) - f(x_i)) / (x_{i+1} - x_i)] \times (x - x_i)$$

Slope of each segment = $\Delta v / \Delta t = (v_2 - v_1) / (t_2 - t_1)$

- Only uses its 2 neighboring data points
- Simple and fastest to compute

Properties & Conditions

- C^0 continuity — function is continuous at knots
- First derivative is DISCONTINUOUS (sharp corners at knots)
- Slope changes abruptly at each data point
- $2n$ unknowns (slope + intercept per segment)
- $2n$ equations (2 endpoints per segment)
- System is exactly determined

Equation Count: n intervals $\times$ 2 endpoint conditions = $2n$ equations | Unknowns: $2n$ (slope & intercept) $\rightarrow$ Exactly determined

Limitation: Produces sharp corners (kinks) at knots — NOT smooth enough for engineering applications.

Linear Spline — Rocket Velocity Example

Problem: Find the velocity at $t = 16$ s using linear splines.

Step-by-Step Solution

Step 1 — Identify interval:

$t = 16$ lies between $t_3 = 15$ and $t_4 = 20$.
Use the spline on $[15, 20]$.

Step 2 — Apply formula:

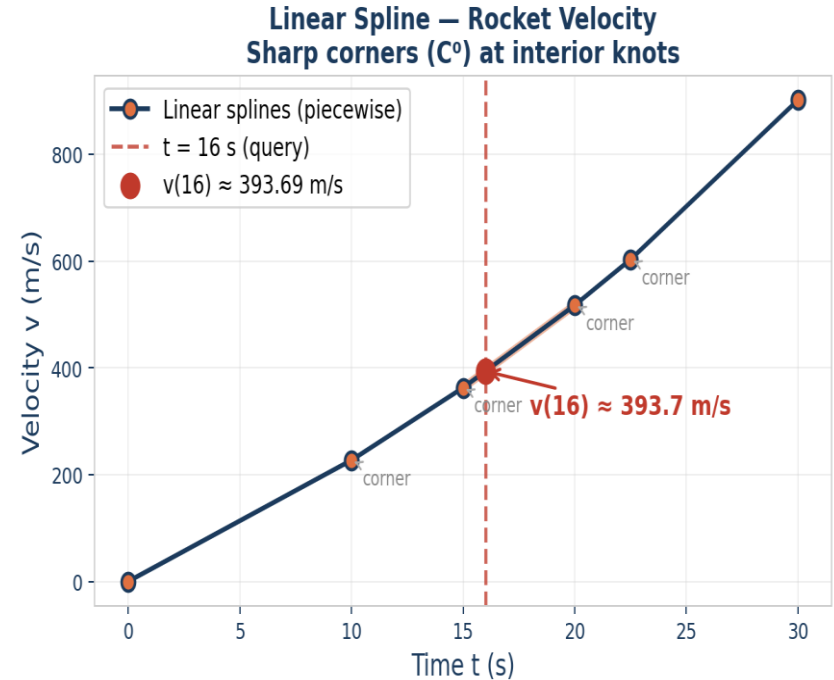
$$f(t) = f(15) + [(f(20)-f(15))/(20-15)] \times (t-15)$$

Step 3 — Substitute:

$$\begin{aligned} &= 362.78 + [(517.35-362.78)/5] \times (t-15) \\ &= 362.78 + 30.914 \times (t-15) \end{aligned}$$

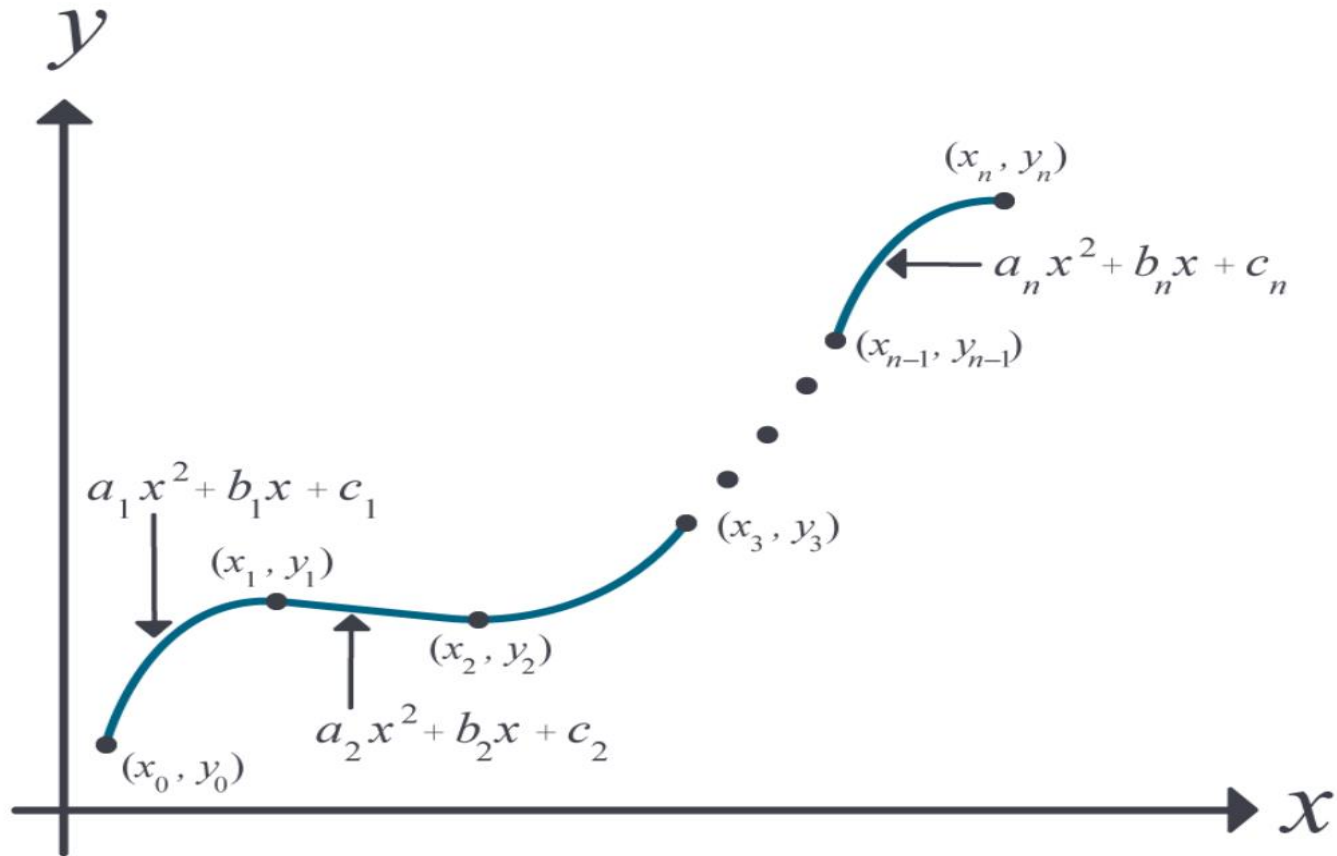
Step 4 — Evaluate at $t=16$:

$$= 362.78 + 30.914 \times 1 = 393.69 \text{ m/s}$$



Result: $v(16) \approx 393.69$ m/s (Linear Spline)

Quadratic Spline Interpolation — Theory & Conditions



Quadratic Spline Interpolation — Theory & Conditions

Concept: Use a quadratic polynomial $v_i(t) = a_i t^2 + b_i t + c_i$ in each sub-interval. C^1 smooth — first derivative is continuous.

Condition 1

Each Spline Passes Through Both Endpoints

For each interval $[t_i, t_{i+1}]$:

$$a_i t_i^2 + b_i t_i + c_i = v_i$$

$$a_i t_{i+1}^2 + b_i t_{i+1} + c_i = v_{i+1}$$

→ **2n equations**

Condition 2

1st Derivative Continuous at Interior Knots

At each interior knot t_i

($i=1, \dots, n-1$):

$$2a_i t_i + b_i = 2a_{i+1} t_i + b_{i+1}$$

(slopes must match)

→ **(n-1) equations**

Condition 3

One Boundary Condition (extra)

Need 1 more equation to close the system.

Commonly: set $a_1 = 0$ or $a_n = 0$ (first and last spline has no curvature → linear).

Others rightly base it on which interval is smaller, $[x_0, x_1]$ or $[x_{n-1}, x_n]$

→ **1 equation**

Total: $2n + (n-1) + 1 = 3n$ equations = $3n$ unknowns (a_i, b_i, c_i) → System exactly determined

Quadratic Spline — Matrix System Setup (Rocket Data, n=5)

5 intervals $\rightarrow$ 15 unknowns ($a_1 \dots a_5, b_1 \dots b_5, c_1 \dots c_5$) $\rightarrow$ 15 $\times$ 15 linear system $[A]\{x\} = \{b\}$

Equations by Type

Endpoint conditions (10 eqns):

Spline 1 at t=0: $c_1 = 0$

Spline 1 at t=10: $100a_1 + 10b_1 + c_1 = 227.04$

Spline 2 at t=10: $100a_2 + 10b_2 + c_2 = 227.04$

Spline 2 at t=15: $225a_2 + 15b_2 + c_2 = 362.78$

Spline 3 at t=15: $225a_3 + 15b_3 + c_3 = 362.78$

Spline 3 at t=20: $400a_3 + 20b_3 + c_3 = 517.35$

Spline 4 at t=20: $400a_4 + 20b_4 + c_4 = 517.35$

Spline 4 at t=22.5: $506.25a_4 + 22.5b_4 + c_4 = 602.97$

Spline 5 at t=22.5: $506.25a_5 + 22.5b_5 + c_5 = 602.97$

Spline 5 at t=30: $900a_5 + 30b_5 + c_5 = 901.67$

Derivative conditions (4 eqns):

$$20a_1 + b_1 = 20a_2 + b_2$$

$$30a_2 + b_2 = 30a_3 + b_3$$

$$40a_3 + b_3 = 40a_4 + b_4$$

$$45a_4 + b_4 = 45a_5 + b_5$$

Boundary (1 eqn):

$$a_5 = 0$$

Quadratic Spline — Matrix System Setup (Rocket Data, n=5)

5 intervals $\rightarrow$ 15 unknowns ($a_1 \dots a_5, b_1 \dots b_5, c_1 \dots c_5$) $\rightarrow$ 15 $\times$ 15 linear system $[A]\{x\} = \{b\}$

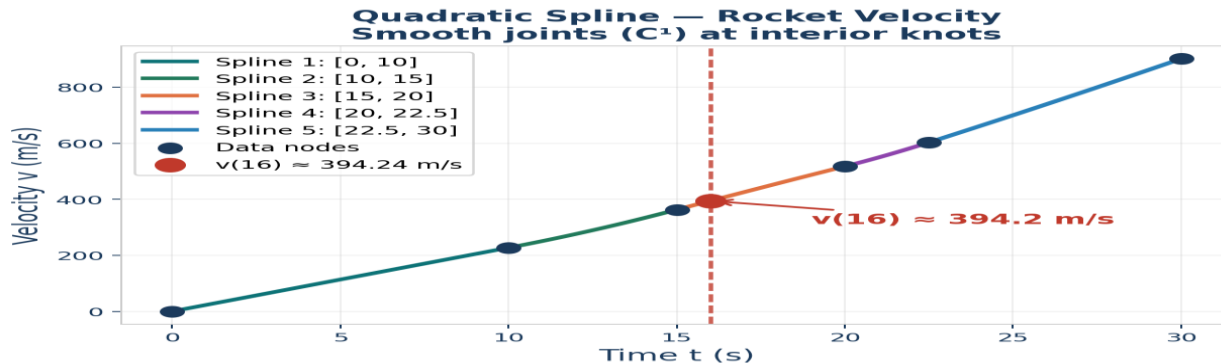
$$\begin{bmatrix}
 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 100 & 10 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 100 & 10 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 225 & 15 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 225 & 15 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 400 & 20 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 400 & 20 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 506.25 & 22.5 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 506.25 & 22.5 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 900 & 30 & 1 \\
 20 & 1 & 0 & -20 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 30 & 1 & 0 & -30 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 40 & 1 & 0 & -40 & -1 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 45 & 1 & 0 & -45 & -1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0
 \end{bmatrix}
 =
 \begin{bmatrix}
 a_1 \\
 b_1 \\
 c_1 \\
 a_2 \\
 b_2 \\
 c_2 \\
 a_3 \\
 b_3 \\
 c_3 \\
 a_4 \\
 b_4 \\
 c_4 \\
 a_5 \\
 b_5 \\
 c_5
 \end{bmatrix}
 \begin{bmatrix}
 0 \\
 227.04 \\
 227.04 \\
 362.78 \\
 362.78 \\
 517.35 \\
 517.35 \\
 602.97 \\
 602.97 \\
 901.67 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0
 \end{bmatrix}$$

Quadratic Spline — Solved Coefficients & Spline Expressions

After solving the 15×15 system, the coefficients for each spline $v_i(t) = a_i t^2 + b_i t + c_i$ are:

Spline	Interval	a_i	b_i	c_i	Expression $v(t) =$
1	[0, 10]	0.00000	22.704	0.00	$22.704t$
2	[10, 15]	0.8888	4.928	88.88	$0.8888t^2 + 4.928t + 88.88$
3	[15, 20]	-0.1356	35.66	-141.61	$-0.1356t^2 + 35.66t - 141.61$
4	[20, 22.5]	1.6048	-33.956	554.55	$1.6048t^2 - 33.956t + 554.55$
5	[22.5, 30]	0.20889	28.86	-152.13	$0.20889t^2 + 28.86t - 152.13$

↑ Row 3 highlighted — $t = 16$ falls in [15, 20], so spline 3 is used for the rocket example.



Quadratic Spline — Finding Velocity, Acceleration & Distance

(a) Velocity at $t = 16$ s

Use spline 3: $v(t) = -0.1356t^2 + 35.66t - 141.61$

$$\begin{aligned}v(16) &= -0.1356(256) + 35.66(16) - 141.61 \\&= -34.71 + 570.56 - 141.61 \\&= 392.19 \text{ m/s}\end{aligned}$$

(b) Acceleration at $t = 16$ s

Differentiate spline 3:

$$\begin{aligned}a(t) &= dv/dt = 2(-0.1356)t + 35.66 \\&= -0.2712t + 35.66\end{aligned}$$

$$a(16) = -0.2712(16) + 35.66 = 32.32 \text{ m/s}^2$$

(c) Distance Covered from $t = 11$ s to $t = 16$ s

$t=11$ is in $[10,15]$ and $t=16$ is in $[15,20]$ → split the integral at the knot $t=15$:

$$\begin{aligned}s &= \int_{11}^{16} v(t) dt = \int_{11}^{15} v_2(t) dt + \int_{15}^{16} v_3(t) dt \\&= \int_{11}^{15} (0.8888t^2 + 4.928t + 88.88) dt = [0.8888t^3/3 + 4.928t^2/2 + 88.88t]_{11}^{15} \approx 1,003.9 \text{ m}\end{aligned}$$

$$\int_{15}^{16} (-0.1356t^2 + 35.66t - 141.61) dt = [-0.1356t^3/3 + 35.66t^2/2 - 141.61t]_{15}^{16} \approx 380.0 \text{ m}$$

$$\text{Total distance } s \approx 1,003.9 + 380.0 = 1,383.9 \text{ m}$$

$$v(16) \approx 392.19 \text{ m/s} \quad | \quad a(16) \approx 32.32 \text{ m/s}^2 \quad | \quad \text{Distance } (t=11 \rightarrow 16 \text{ s}) \approx 1,383.9 \text{ m}$$

Cubic Spline Interpolation — Theory & Conditions

Concept: Use a cubic $S_i(x) = a_i + b_i(x-x_i) + c_i(x-x_i)^2 + d_i(x-x_i)^3$ per interval. Achieves C^2 smoothness — the gold standard.

Cond 1 — Interpolation

Each cubic passes through both endpoints of its interval.

$$S_i(x_i) = f(x_i) \text{ and } S_i(x_{i+1}) = f(x_{i+1})$$

→ **2n equations**

Cond 2 — C^1 : 1st Derivative

Adjacent splines have equal first derivatives at interior knots.

$$S_i'(x_{i+1}) = S_{i+1}'(x_{i+1})$$

→ **(n-1) equations**

Cond 3 — C^2 : 2nd Derivative

Adjacent splines have equal second derivatives at interior knots.

$$S_i''(x_{i+1}) = S_{i+1}''(x_{i+1})$$

→ **(n-1) equations**

Cond 4 — Boundary Conds (2)

Natural: $S''(x_0)=0, S''(x_n)=0$

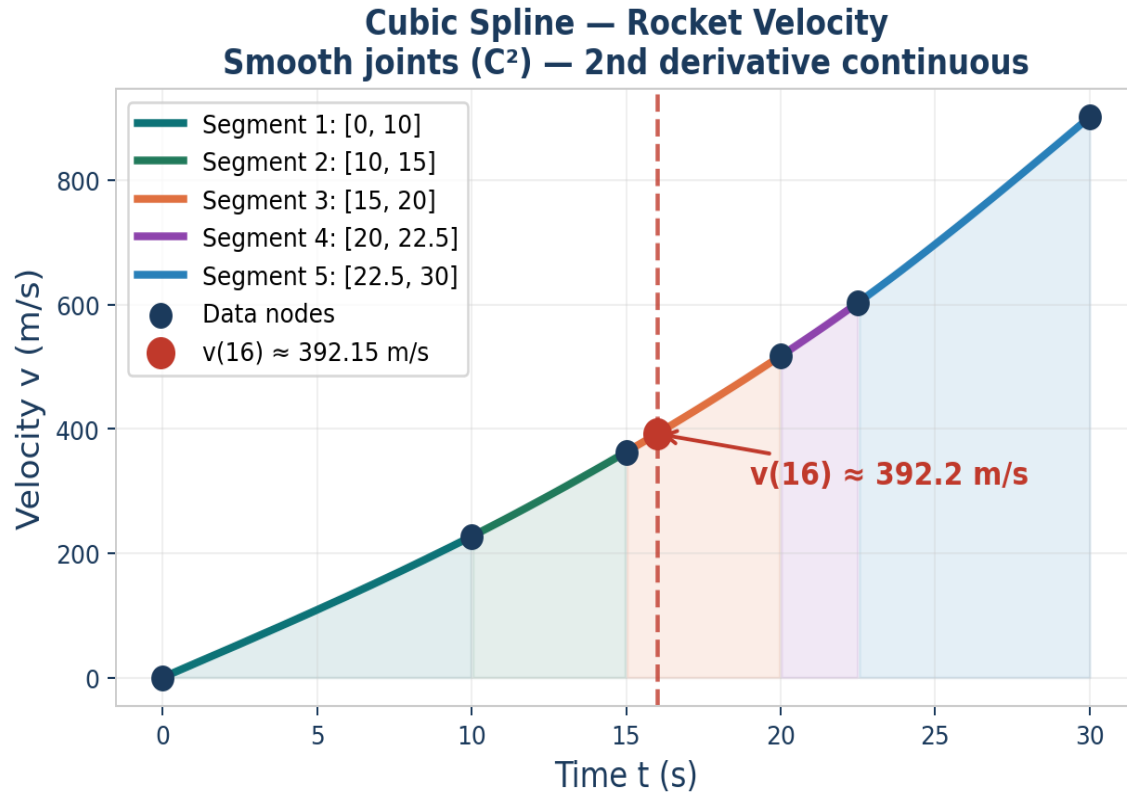
Clamped: $S'(x_0)=f'_0, S'(x_n)=f'_n$

Not-a-Knot: 3rd deriv. cts at x_1 & x_{n-1}

→ **2 equations**

Total: $2n+(n-1)+(n-1)+2 = 4n$ equations = $4n$ unknowns → Fully determined | Solve with LU algorithm

Cubic Spline — Rocket Velocity (Natural Boundary Conditions)



Cubic Spline Details

- $4n = 20$ unknowns for $n=5$ intervals
- Natural BC: $S''(0)=0$ and $S''(30)=0$
- Segments coloured individually
- 2nd derivative is continuous
- No oscillation at all between knots
- Most accurate of the three types

Compare accuracy:

- Linear: $v(16) \approx 393.69$ m/s
- Quadratic: $v(16) \approx 392.19$ m/s
- Cubic: $v(16) \approx 392.06$ m/s

Cubic spline gives the smoothest curve and highest accuracy for the same data set.

Summary Comparison — Linear, Quadratic & Cubic Splines

Property	Linear	Quadratic	Cubic
Degree of polynomial	1	2	3
Unknowns per interval	2	3	4
Total unknowns (n intv.)	2n	3n	4n
Continuity class	C^0	C^1	C^2
1st derivative at knots	✗ Discontinuous	✓ Continuous	✓ Continuous
2nd derivative at knots	✗ Discontinuous	✗ Discontinuous	✓ Continuous
Boundary conditions needed	0	1	2
v(16) rocket example	393.69 m/s	392.19 m/s	392.06 m/s
Typical use	Quick estimates	Intermediate accuracy	Engineering applications

Key Takeaway: Cubic splines deliver C^2 smoothness and highest accuracy, making them the industry standard in engineering, computer graphics, CAD, and numerical analysis.